

Enabling Face Tracking in an Affordable Social Robot Using a Computer Vision Camera

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Abstract—Social robotics has been explored as a tool to support social interactions in various applications of human well-being. In therapies of children with Autism Spectrum Disorder (ASD) social robots have been included as motivational tools to perform activities within the therapeutic session. This work presents a contribution to the natural behaviours of the CASTOR robot, an open-source robot for assisted therapies of children with ASD. The project proposal involves using a camera with computer vision and implementing face recognition to improve the movements of the robot's neck and eyes. The tests were carried out in two phases; first, validating the functioning of the eye-contact tracking; and then, on the neck movement. The resulting improvements provided more natural and engaging gaze behaviour, where the responsiveness of the system obtained an accuracy of 96.11% in the eyes and an RMSE of 3.62° of the tracking angle for the neck. Future work will evaluate user perception of the natural gaze system in a social interaction activity using standardized questionnaires.

Index Terms—Face Tracking; Human-Robot Interaction; Sensor-based Control; Social behaviour.

I. INTRODUCTION

Socially Assistive Robotics (SAR) has gained increasing attention as an interdisciplinary field that merges robotics, health, and human-computer interaction to support social interactions [1]. These robots aim to promote user engagement, motivation, and social-emotional development, particularly in healthcare, education, and therapeutic contexts [2].

Autism Spectrum Disorder (ASD) is characterized by challenges in social communication, sensory processing, and the expression and recognition of nonverbal behaviours such as gaze, gestures, and facial expressions [3]. Children with ASD often benefit from structured and predictable social environments, and SARs have been explored as promising tools to achieve these interactions in engaging ways [4].

Recent implementations of SARs in ASD interventions have demonstrated positive outcomes in supporting communication skills, emotional regulation, and joint attention [5].

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 – "CPID 2.2", FAPES/ CNPq/Decit – PPSUS 17/2024, 2023–XDHHW), CNPq (306323/2024-9, 404111/2023-8, and 402739/2024-8), Finep (Tec assistiva 2 - 2132/22), and IEEE RAS SPARK.

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Robots like NAO (Aldebaran Robotics, France), KASPAR (Hertfordshire University, United Kingdom), and QTrobot (LuxAI, Luxemburg) have been used to facilitate therapeutic sessions, especially when their behaviours mimic human social cues [6]. Among these cues, eye contact and gaze direction are particularly analyzed as essential for real-world social interactions [7]. Natural gaze behaviour, where the robot can appear to attend and respond to a person, has been shown to increase trust, engagement, and perceived intelligence in human-robot interactions [8].

Some robots with gaze natural behaviours are Pepper and Furhat [9], implemented in social scenarios, evaluating the influence of gaze patterns in perception and comparing two complex robotic platforms. However, most of the commercially available SARs are expensive and rely on complex sensor suites, limiting their scalability in low-cost, designed robots such as the Shadow robot [10].

In this context, the CASTOR robot has addressed these limitations through innovative design, integrating flexible soft robotics and elastic actuators [11]. The device was projected to provide natural Human-Robot Interaction (HRI) and support therapeutic activities with neurodiverse children [12]. Its flexibility, huggable structure, and shock-absorbing mechanisms ensure the robot's durability and users' safety [13]. In addition, the robot is built using 3D-printed parts, actuators, and microcontrollers. Previous interactions of CASTOR focused on basic interactive behaviours and activities during speech therapy sessions. However, the inclusion of natural gaze mechanisms, such as head turning and eye movement in response to a human's position, remains an open challenge in its development [14].

In recent years, advances in computer vision (CV) have made it feasible to integrate camera-based perception systems into robotic platforms for face detection, gaze estimation, and emotion recognition [15]. Low-cost RGB cameras, combined with open-source CV libraries, allow the implementation of robust interaction mechanisms without the need for expensive sensors (e.g., Light Detection and Ranging (LiDAR)) [16]. In SAR, CV enables robots to dynamically adapt their behaviours based on user position and facial expressions, improving interaction flow and responsiveness.

Previous works have integrated CV into SARs to enable behaviours such as gaze following, particularly in interactions with children with ASD [16]. For example, the QTrobot uses a depth camera and facial recognition to track gaze direction and body tracking, showing improved engagement levels in children [17]. Similarly, the Zeno robot has been

used to model joint attention behaviours through head orientation based on the user's location [18]. Likewise, the NAO robot has been used with Kinect cameras to identify face characteristics and identify variables and children's behaviours during robot-assisted therapies [19]. Lastly, low-cost robots like ONO have also implemented external camera sensors to identify Fields of View (FoV) and performance in joint attention therapy [20]. These studies indicate that robotic behaviours driven by perceptual cues significantly improve user experience, attention, and therapeutic engagement.

This work presents the integration of a computer vision-based system for controlling natural gaze behaviours (neck and eye movement) on the CASTOR robot. The objective is to improve the robot's gaze and facial direction through real-time face tracking using an AI camera, evaluating the Root Mean Square Error (RMSE) and the standard deviation of the eye section and motor angle. The system was validated in two phases: first, testing the eye contact, and then the neck movement toward the user. This work contributes to the advancement of the capabilities of the CASTOR platform for future implementations in therapies, presenting the methodology describing the robot used, the system and experimental procedure, the results for each subsystem (eyes and neck), discussion, conclusions, and future work.

II. METHODOLOGY

The methodology was designed to assess the functional performance of individual subsystems to identify natural social interactions by using computer vision. This section is organized into four main components: (A) the robotic platform and its architecture; (B) the materials used in the actuation modules, including the Huskylens AI camera, the LCDs eyes, and a Dynamixel motor for neck movement; (C) the experimental procedure, divided into both subsystems validation with a neurotypical adult volunteer; and (D) the data analysis, which includes technical metrics of detection (i.e., accuracy and error rate).

A. Robotic Platform

The CASTOR robot is an affordable, open-source platform designed for therapies of children with ASD. To ensure the safety of users, it has elastic actuators and bioinspired tendons. The robot is equipped with seven motors (Dynamixel MX/AX, Robotis, France), five servomotors (MG996, Towerpro, Taiwan), and a passive damping system resulting in 14 degrees of freedom (DoF). Regarding the functionalities of interest, the system includes neck movement and Thin-Film Transistor (TFT) Liquid Crystal Display (LCD) eyes control, which allows the integration of the computer vision system for face detection and natural gaze toward users. Besides, CASTOR is a robust 3D-printed modular device, easy to build, modify, and repair. In this work the robot structure was adapted to add the camera directly on the robot's head. In addition, a responsive antenna was added in Polyethylene

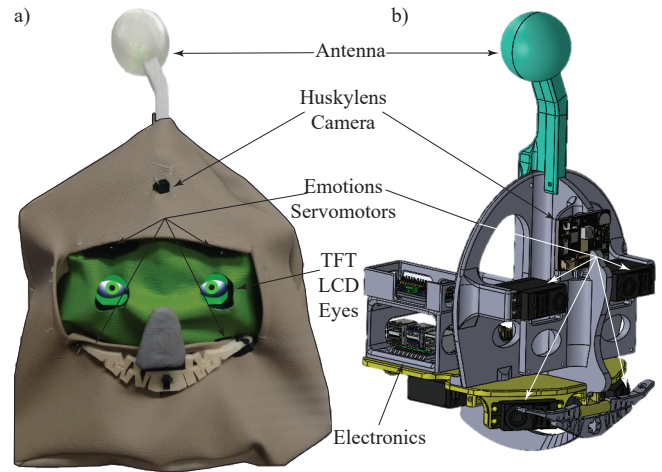


Fig. 1: Robotic platform. a) The robot used and b) the head's CAD of the modifications of the 3D printed part.

terephthalate glycol (PETG) with a color feedback LED, as presented in Figure 1a. In both subsystems, the nonverbal feedback includes that when the camera recognizes someone in its FoV, the antenna turns green and the robot smiles. The CAD of the head modification is shown in Figure 1b.

B. Materials

To enable face tracking, it was used the Huskylens AI camera module (DFRobot, China), which is an easy-to-use, compact, and cost-effective AI vision sensor with six built-in functions: face recognition, object tracking, object recognition, line tracking, color recognition, and qr code recognition. The Huskylens offers a built-in processor (Kendryte K210) with asynchronous and Inter-Integrated Circuit (I²C) synchronous serial communication capabilities, making it suitable for real-time feedback control without requiring high computational power and with established communication protocols. The camera was mounted on the head of the robot and angled to match the typical height of child users during tabletop interaction, as shown in Figure 2. The camera has a resolution of 320 x 240 pixels, which is used for the boundaries of the nine detection zones, selecting the face recognition mode. The camera can detect faces at different distances until 3 m.

The CASTOR robot uses animated digital eyes rendered on two TFT displays (ST7789) driven by SPI. These displays are controlled by the Adafruit Animated Eyes Bonnet for Raspberry Pi, which renders smooth, responsive, and lifelike animations. These eyes provide visual expressiveness and support configurable eye color and behaviours. The eye display simulates nine distinct visual zones, defined based on face positions detected by the Huskylens camera, as shown in Figure 2a. The head orientation uses the Dynamixel AX-12 servo motor in the neck, connected to the Raspberry Pi through USB and using a U2D2 controller. The motor was responsible for the yaw movement (left-right rotation) of the robot's head to produce a coherent and natural head direction. The neck mechanism was adapted to prioritize smooth and

human-like motion toward the user, and the motor angle was used for the tracking validation.

Both actuators and the AI camera used the Robot Operating System (ROS) to communicate the position and activate the zone or angle, respectively. The algorithm feeds the actuators' position according to the user's face position in the camera data, the HuskyLens returns a block with essential information of ID, *x*-and-*y*-axis positions, and height and weight of the detected face. With this information, if the ID is > 1 , the speaker reproduces a greeting mp3 file with the saved name. With the position, the *x* and *y* is sent to the eyes node and a 9-option selector is used to generate the eyes behaviour, and the *x* is received in the neck's node to move horizontally. Finally, the height and weight is used to select the nearest face to follow, the code for the algorithm is in the CASTOR robot repository¹.

C. Experimental Procedure

The procedure was divided into two steps, the evaluation of the natural behaviour of 1) the eyes, and 2) the neck motor, controlled by the face detection of the camera. All experiments were conducted in a controlled indoor environment, with consistent lighting and minimal background noise, to ensure the camera's performance. A researcher manually guided a neurotypical adult volunteer across predefined spatial zones to simulate the positions during user interaction, indicating the 30-second mark.

1) *Eyes tracking subsystem*: To evaluate the responsiveness and accuracy of the eye movement, the robot was programmed to track nine distinct facial positions distributed in a 3x3 grid covering the horizontal and vertical FoV of the camera, as shown in Figure 2a. A volunteer stood at a fixed distance ($\sim 1\text{m}$) and slowly transitioned between these positions, considering the pixel resolution of the Huskylens camera. Each position was held for 30 seconds, and the test was repeated five times. The eye display output was saved and compared with the intended gaze zone to determine the tracking accuracy.

2) *Neck motor subsystem*: To validate the robot's head rotation, the system was tested across 10 angular positions, spaced at 10° intervals, ranging from -45° to $+45^\circ$. This subsystem covers a greater horizontal range, but is limited in the vertical gazing range. The volunteer moved to each target location and remained still for 30 seconds, and the test was repeated five times. The resulting motor angle was saved, and the deviation from the expected angle was calculated. The trials were carried out with the setup presented in Figure 2b.

D. Data Analysis

All data were recorded using Python scripts integrated with ROS at 54 Hz, and the results were analyzed by comparing the measured and expected data. Accuracy and error metrics were extracted for each subsystem as follows:

- *Eye's zone*: For the eyes, accuracy was defined as the percentage of correct responses in which the robot directed its gaze to the correct zone within the 30-second segment, which is when the direction of the displayed

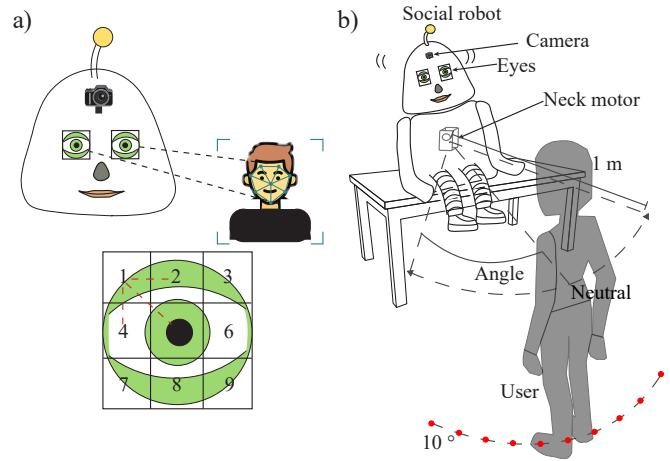


Fig. 2: Validation setup for both subsystems functioning. a) The eye-tracking representation, dividing the nine eye areas. b) The motor movement toward the user, evaluating the neck's motor angles.

eye matched the expected one from the 3x3 grid. The correct responses were calculated for each trial, and the mean accuracy was used to analyze the system's performance. The error was set based on the Euclidean distance between adjacent regions with discrete values for each zone and normalized to analyze the incorrect detection across zones (e.g., Figure 2a shows that for the zone 1 the assigned error was one when the incorrect detected area was 2 or 4, two when it was 5, three when it was 3 or 7, four when it was 8 or 6, five when it was 9, and six when the camera losses the user face).

- *Motor angle*: In the case of the neck position, the data was compared with the expected angle marked in the floor as represented in Figure 2b. The measured angle was defined as the maintained angle in the set position of the motor during the 30 seconds and compared with the steps from -45° to $+45^\circ$. The estimated angle was compared to obtain the RMSE of the user position across the five trials, and at the end across all 10 possible angles.

III. RESULTS

The subsystems were tested and analyzed to improve the natural gaze behaviour of the robot during interactions with users, a summary is presented in Table I. Figure 3 presents the comparison between the real target positions and the mean measured output from the robot's eyes, including the standard deviation margin. In addition, with these results, the error was calculated with the distance between corresponding regions.

The system demonstrated high consistency in identifying face position and maintaining gaze direction across the tested areas. The mean error relative to the ground truth remained below one unit in most segments, indicating accurate detection and reliable mapping between face position and eye direction. Occasional spikes in error were observed

TABLE I: Performance results for both subsystems for each trial, the average, and its standard deviation. The eyes subsystem in % and the neck in $^{\circ}$.

Eyes						
ACC1	ACC2	ACC3	ACC4	ACC5	Ave	SD
91,67	96,48	93,89	98,89	99,63	96,11	3,00
Neck						
MAE1	MAE2	MAE3	MAE4	MAE5	Ave	SD
1,58	2,45	2,45	2,78	3,38	2,53	0,58
RMSE1	RMSE2	RMSE3	RMSE4	RMSE5	Ave	SD
2,51	3,65	3,60	3,89	4,46	3,62	0,63

during transitions or at the boundaries between adjacent zones, likely due to partial facial movement toward the next region. Overall, the subsystem achieved an average tracking accuracy of $96.11\% \pm 3.00\%$, which is suitable for HRI applications [14].

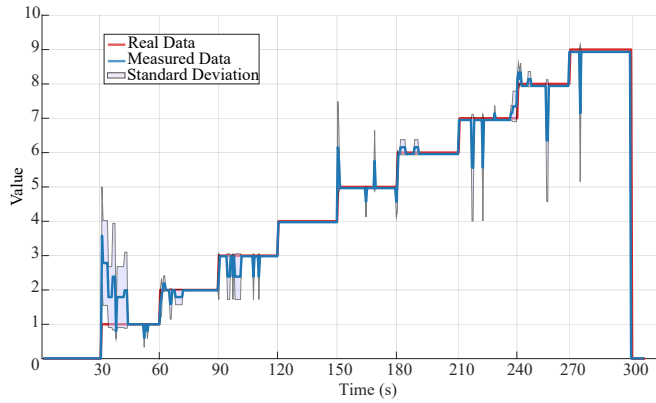


Fig. 3: Comparison between real target positions and the mean measured eye tracking subsystem output of the five trials with the standard deviation.

The motor control performance, illustrated in Figure 4, showed a mean RMSE of $3.62^{\circ} \pm 0.63^{\circ}$ across the 90° FoV. Interestingly, the system exhibited better accuracy at the extremes of the FoV, while higher variation was observed in the central range (-10° to 10°). Across individual steps, increased RMSE was noted near the boundaries, also attributed to transitions between discrete angle steps. Despite these fluctuations, the overall performance was acceptable, with RMSE consistently remaining below $\pm 5^{\circ}$ in the five conducted trials. Lastly, examples of the two subsystems interactions are illustrated in Figure 5, the subsystem of the eyes is following the user in position 4 (center-left), and the subsystem of the neck is directed toward the user at -35° .

IV. DISCUSSION

The results obtained from the subsystems demonstrate its ability to dynamically and accurately identify facial positions across the defined 3×3 grid, achieving a high average accuracy of 96.11% and minimal mean error in most regions. This performance is relevant for real-time HRIs, where accurate gaze tracking and low error rates are essential to ensure natural and responsive behaviours. When compared to other approaches, such as the facial pattern classification technique

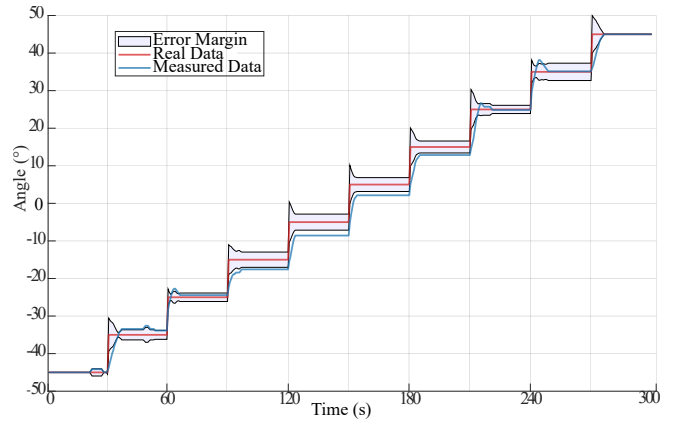


Fig. 4: Comparison of the robot's neck position and the real target angle with the calculated RMSE.

in five zones presented in [14], which achieved up to 95% accuracy for vertical (top and bottom) and 93% for horizontal (left and right) gaze zones using a neural network, the current implementation offers 96% accuracy while operating at a finer-grained spatial resolution (nine zones). Additionally, while Nurmaini *et al.* [14] focused on navigation decisions in mobile robots, the present work emphasizes the control of socially natural outputs, where the continuity and stability of the gaze signal can affect users' perception and experience. The error rate lower than one unit in all the segments, and the smooth transitions between zones observed in this system further highlight the robustness of the approach in dynamic settings for facial-guided robotic behaviour control.

The higher RMSE in the central interval (-10° to 10°) of the neck subsystem is explained by the Huskylens coordinate system, where the image center (160,120) is the intersection of the x and y axes dividing the frame into four regions. In the control code, this is implemented through boundary conditions ($140 \leq x \leq 180$), which overlap with the lateral ranges (≤ 140 and ≥ 180). As a result, when the user's face is near the center, even a ± 1 pixel fluctuation can change the classification between central and lateral zones, producing minimal instability and error in neck positioning. This boundary ambiguity, amplified by the camera's limited 320×240 resolution, explains the increased RMSE in the neutral range. Future improvements could redefine region thresholds to avoid overlap, apply hysteresis margins, or use probabilistic assignment to stabilize classification near the coordinate cross [21], [22].

Referring to nonverbal communication and behaviour during robotic implementations, other studies have examined the impact of robot functionalities on user perception. Janeczko *et al.* [9], for instance, compared the Furhat and Pepper robots in a conversational interaction. They concluded that differences in non-verbal behaviours, such as gestures and face-tracking, significantly influence user perception. Thus, not only behaviour, but also robot functionalities and appearance play a crucial role in shaping the user experience and, consequently, the perception of social intelligence.

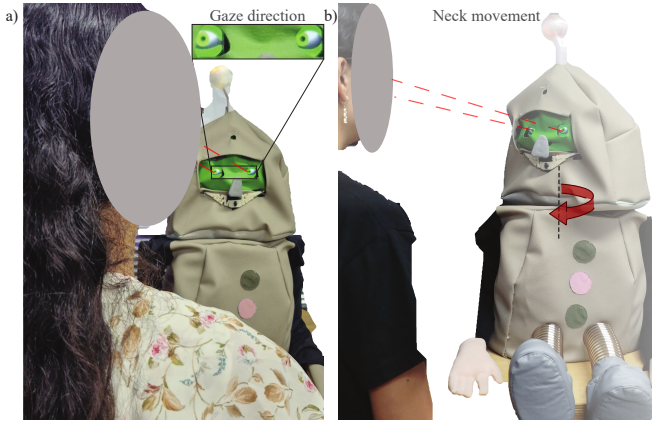


Fig. 5: User detection with the CV system and controlling the gaze direction in a) the eyes subsystem, and b) the neck position subsystem.

TABLE II: Results comparison with other related studies, in method, sensor, and robotic application.

Cite	Method	Main Result	Sensor/ Application
[14]	Facial pattern classification in five zones	95% accuracy for vertical (top and bottom) and 93% for horizontal (left and right) gaze zones	RGB standard camera/ Robotic movement
[25]	Face tracking with a CNN	99%, 95%, and 97% of precision, recall, and F-score	Kinect RGBD camera/ UR5e
[26]	Gaze estimation with a CNN to control gaze behaviours	AUC of 0.92 and an angular error of 13°	High definition camera/ Furhat robot
This	Face tracking to work control gaze and neck direction	Gaze tracking accuracy of 96.11% and neck direction RMSE of 3.62°	AI Huskylens camera/ CASTOR robot

Other works have explored the effect of human-inspired gaze behaviour on user experience, including neck movement during interactions using the Pepper robot. Koller *et al.* [23] concluded that while high mutual gaze can affect interaction duration, it is not always beneficial, and future work should include randomized gaze control in a way that maintains natural engagement with the user. In this way, providing human-inspired natural behaviours, especially in social robotics, is important to maintain the user experience. This approach shows improvements to the gaze behaviours of the CASTOR robot by adding a camera system to control two subsystems. Moreover, multisensor systems are more robust for sensor-based decision-making, for which the robot's architecture is projected to integrate additional sensing-type modules, such as proximity and tactile sensors [24]. A comparison with similar works and results is presented in Table II.

Some limitations were identified during the static tracking validation. Before applying the system in ASD therapies, it is necessary to conduct dynamic tests under non-controlled conditions to better assess its performance. Moreover, several assumptions were made regarding the clinical setup, including environmental lighting, the number of participants in therapy sessions, and users' reactions to specific behaviors.

Finally, the medium angular error within the range of -10° to 10° requires improvement to enable reliable neck movement tracking and to integrate both subsystems, thereby combining gaze and face direction for real-world HRI scenarios.

V. CONCLUSION AND FUTURE WORK

This work presents the integration of nonverbal social behaviour into an affordable social robot, contributing to more natural and engaging HRIs. This was achieved by implementing a camera-based sensor to control two subsystems, generating human-inspired face tracking behaviours. The face detection module enabled real-time control of the eyes with an accuracy of 96.11%, and of the neck motor with an RMSE of 3.62° , demonstrating the reliability and responsiveness of the system. These results indicate the possibility of incorporating gaze behaviours into affordable robotic platforms, which is particularly relevant for applications in social, educational, and therapeutic contexts.

Future work will focus on expanding the robot's sensory capabilities by integrating additional inputs such as tactile and proximity sensors. Additionally, it is planned to evaluate the impact of these nonverbal behaviours in real-world interactions by evaluating and quantifying the social intelligence perception with and without this functionality in the CASTOR robot. More variations should be considered, such as dynamic detection, multi-face tracking, and at different distances. In this way, future implementations will be conducted in children with ASD during therapy, assessing the performance of the system in real-world settings and the effects on the children's attention toward the robot.

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